

Morphophonology

Class 3: Case studies



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Three mechanisms

- Prosodic boundaries
- Cyclic derivation
- Transderivational correspondence

Does one of these suffice? Do we need more than one?



German glottal stops

German glottal stops: reminder

(McCarthy and Prince 1993)

(a)	/juʁ-iv-ən/	[juʁivən]	'adjudicate'
	/pʁɔfɛsoːʁ/	[pʁɔfɛsoːɐ̯]	'professor'
	/holand/	[holant]	'Holland'
	/tsvai-bainiç/	[tsvaibainiç]	'two-legged'
	/egal/	[egal]	'indifferent'
(b)	/fɛʁ-ivən/	[fɛʁʔivən]	'lose one's way' (verirren)
	/auf-ɛsən/	[aufʔɛsən]	'eat up' (aufessen)
	/tsol-amt/	[tsolʔamt]	'customs house' (Zollamt)
	/ʁaib-aizən/	[ʁaipʔaizən]	'grater' (Reibeisen)
	/bɛʁg-ab/	[bɛʁkʔap]	'downhill' (Bergab)

Prosodic boundaries: reminder

- McCarthy and Prince (1993) use Align-L

/aʊf-ɛs-ən/	Onset	Align-L	Dep(?)
a. [aʊf.ʔ ɛ.sən]		*!***	*
b. [aʊ.f ɛ.sən]		*!***	
c. [aʊf].[ɛ.sən]	*!		
d. [aʊf].[ʔ ɛ.sən]		*?	*
e. [aʊ].[f ɛ.sən]		*	

Some options

- Fine-grained Align violations ($? < f$)
 - Align-R for the prefix?
 - $\text{Align}(\textit{auf}, R, \text{PrWd}, R)$
 - How to generalize to $\text{Align}(\text{prefix}, R, \text{PrWd}, R)$?
 - Add morphological affiliations to surface representations?
- (or is this problem fatal for the Align approach?)

Cyclic derivation

- Schematically:

Cycle 1	/ɛsən/
Glottal epenthesis	ʔɛsən
Cycle 2	/aʊf+ʔɛsən/
Glottal epenthesis	—
Output	aʊfʔɛsən

- Glottal epenthesis applies initially in first cycle, and the ʔ is preserved in later cycles

A little more precisely: cycle 1

- ? is required for "vowel initial words"
 - /amt/ → [ʔamt]
- ? is may appear in careful speech in some dialects/idiolects /V_ 'V (Theater, Oase)
 - See Alber (2001) for details
- ? is not permitted elsewhere
 - [matʁiks], [lakʁits], *[maʔʁits], *[maʁʔits]
 - *[maʔ]



Restating the distribution of ?

- Glottal stops are not permitted, except where required by...
 - $*\#V$
 - $*['V$ ("ONSET" in initial and stressed syls)
- Ranking
 - $*\#V, *['V, \text{Dep(place)} \gg *? \gg \text{Dep(?), Max(?)}$

#V,['V, Dep(place) >> *? >> Dep(seg), Max(?)

/amt/	*#V	Dep(place)	*?	Dep(seg)	Max(?)
a.  amt			*	*	
b. amt	*!				
c. tamt		*!		*	

/ma?/	*#V	Dep(place)	*?	Dep(seg)	Max(seg)
a.  ma					*
b. ma?			*!		

/maʔo/	*#V	Dep(place)	*?	Dep(seg)	Max(seg)
a.  maʔo					*
b. maʔo			*!		

Cyclic derivation: *auf-essen*

- Cycle 1

/ɛsən/	*#V	*?	Dep(?)	Max(?)
a.  ?ɛsən		*	*	
b. ɛsən	*!			

- Cycle 2: can't be same ranking

/auf+?ɛsən/	*#V	*?	Dep(seg)	Max(seg)
a.  ?auf?ɛsən		**!		
b.  ?aufɛsən		*		*
c. auf?ɛsən	*!	*		
d. aufɛsən	*!			*

Solution: later cycles promote Max(?)

/auf+ʔəsən/	*#V	Max(seg)	*ʔ	Dep(seg)
a.  ʔaufʔəsən			**	
b. ʔaufəsən		*!	*	
c. aufʔəsən	*!		*	
d. aufəsən	*!	*!		

- Glottal stops inserted in cycle 1 are preserved in cycle 2

A recurring risk: contrasts in affixes

- Affixes added in cycle 2 are evaluated with the ranking of cycle 2
- If Faithfulness (Max(?)) is higher ranked, these affixes could have post-consonantal ?

/ybəʋ+X/	*#V	Max(seg)	*?	Dep(seg)
a.  ybəʋ...				
b. ybʔəʋ...			*	*

/ybʔəʋ+X/	*#V	Max(seg)	*?	Dep(seg)
a. ybəʋ...		*!		
b.  ybʔəʋ...			*	



Some options

- Additional reference to prosody (see Catalan)
- Add a mechanism that bans /?/ in lexical entries
 - Morpheme Structure Constraints (next week)
 - Could address the problem for German, but can't preclude the possibility of other languages with this pattern
- Other options?

(or is this problem fatal for the cyclicity approach?)



Transderivational faithfulness

- Approach: *auf*-[?]essen has a glottal stop because [?]essen does
- BD-Max(?)
 - Glottal stops in the base must have a correspondent in the derived form

Transderivational faithfulness: BD-Max(?)

/ɛsən/	BD-Max (seg)	*#V	*?	IO-Dep (seg)	IO-Max (seg)
a.  ʔɛsən			*	*	
b. ɛsən		*!			

/auf+ɛsən/	BD-Max (seg)	*#V	*?	IO-Dep (seg)	IO-Max (seg)
a.  ʔaufʔɛsən			**	**	
b. ʔaufɛsən	*!		*	*	
c. aufʔɛsən		*!	*	*	
d. aufɛsən	*!	*			



Features of this account

- Only favors phonotactically "unexpected" ? stem-initially, because this is where bases may have ?
- Regular phonotactics enforced in prefix
- ? only expected in cases where there is an independently occurring (free-standing) base
 - Glottal stop epenthesis overapplies in *auf-[?]essen* due to faithfulness to *[?]essen*



An Achilles heel: bound forms

- Bound stems should not show epenthesis
 - e.g., hypothetical *auf-effen* with no corresponding **effen*
- German does have bound roots
 - *Ver-nunft* 'reason' (**Nunft*), *ver-hunzen* 'botch' (**hunzen*)
- Vowel-initial bound roots?



Prefixed V-initial stems without bases?

- A few prefixed verbs without corresponding unprefixated verbs, but related nouns/adjectives

beabsichtigen	'intend'	*absichtigen	Absicht (n.)
umarmen	'embrace'	*armen	Arm (n.)
veranstalten	'arrange'	*anstalten	Anstalt (n.)
beunruhigen	'upset'	*unruhigen	unruhig (adj.)
erobern	'conquer'	*obern	obere (adj.)

- These require BD-Max(?) for cross-category derivation (always obeyed in German)
- Important question: which is the base, which derived?

Prefixed V-initial stems without bases?

- Cases with no (universally used) base at all
 - beobachten* 'observe' dial. *obacht!*
 - beömmeln* 'amuse oneself' dial. *ömmeln(?)*
- These particular examples involve /V'V/, which is anyway a context for ? epenthesis
 - *The[?]ater, Ko[?]ala*
 - Dialect and speech rate/style variability
 - Same variability in *beobachten, beömmeln?*
- Cleaner test: bound bases after C- (*ver-*, *auf-*, ...)



Summary: German ? epenthesis

- All three approaches get in the ballpark, but with wrinkles
 - Prosody approach: what counts as aligned?
 - Cyclic approach: avoiding contrast in affixes?
 - Transderivational approach: bound bases?
- Transderivational approach makes unique predictions, but relevant data is scant

English vowel nasalization



English: *minus* vs. *shyness*

- Raffelsiefen (2005)

mãĩnəs 'minus'

ʃaɪ'nəs 'shyness'

p. 211: "the phonetic contrasts concern several properties: *shyness* and *minus* differ both in the length of the diphthong... and in the amplitude of *n*. Umeda and Coker (1974:5) characterize the phonetics of suffix-initial *n* as 'more consonantal' as opposed to the medial *n* in words like *minus*, which is characterized as 'more vocalic'."



minus vs. *shyness*: prosodic boundaries

- Raffelsiefen (2005, 2007): prosodic difference
 - [minus]_{PrWd}
 - [shy]_{PrWd}[ness]_{PrWd}
- Durational difference due to grammar of timing at prosodic boundaries
- Differences in vowel nasalization and duration of [n] due to prosodic effects on gestural strength and coarticulation
- Open question: is *-ness* really a separate PrWd?



minus vs. *shyness*: cyclic evaluation

- Some property of first cycle must be carried forward to subsequent cycle for /*ʃaɪ-nɛs*/
- Two candidate properties
 - Duration of [*ʃaɪ*]
 - Prosodification of [*ʃaɪ*]
- Raffelsiefen's worry about Ident(duration)
- Prosodification solution approximates prosodic boundaries approach



minus vs. *shyness*: BD faithfulness

- Raffelsiefen (2005) sketches how BD-Ident can be extended to this subtle phonetic difference
- Isolation *shy* [ʃaɪ]: monosyllabic, utterance final
 - Longer vowel duration
- Derived *shy-ness*: BD-Ident(duration)
 - Outranks durational constraints that ordinarily demand shorter duration in a disyllable (*minus*)

Catalan sibilant voicing



Voicing-related phenomena

- Contrast in voicing before a vowel within the word
 - *casa* [kázə] ‘house’ vs. *caça* [kásə] ‘hunting’
- Voicing assimilation before a consonant (across the board)
 - *Oslo* [ózlu], *cas lògic* [kaz lózik] ‘logical case’
- Word-final devoicing
 - *casos* [kázus] ‘cases’ vs. *cas* [kás] ‘cas’



Sibilant voicing across words

- Word-final sibilants become voiced when the following word starts with a vowel:

<i>cas</i>	[kás]	‘case’	<i>cas obert</i>	[kà.z uβért]	‘open case’
<i>cares</i>	[kárəs]	‘faces’	<i>cares amables</i>	[kárəz əmápləs]	‘friendly faces’
<i>mires</i>	[mírəs]	‘you watch’	<i>mires això</i>	[mírəz əʃó]	‘you watch this’

Sibilant voicing in prefixes

Interestingly, sibilant voicing before a vowel also applies when the 's' belongs to a prefix.

desinflar [dəzimflá] 'deflate'

desànim [dəzánim] 'discouragement'

Something that we will have to take into account: resyllabification across words and across prefixes.

cas obert [kà.z uβért], *desànim* [də.zánim]



Constraints

- $*\text{Lar}]_{\sigma}$: no laryngeal features in coda position
- $\text{Ident}(\pm\text{voi})$: input and output must have the same values for voicing
- $\text{Max}(\text{lar})$: every laryngeal feature in the input has a correspondent in the output
- No laryngeal features $\Rightarrow$ no $[\pm\text{voi}]$ feature
 - Delaryngealized 's', represented as 'S'

Voicing contrast

- casa* 'house'

/kázə/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a.  ká.zə				
b. ká.sə			*!	
c. ká.Sə				*

- caça* 'hunting'

/kásə/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. ká.zə			*!	
b.  ká.sə				
c. ká.Sə				*

Cyclic approach: the main idea

- Positions with contrast: phonology outputs [s],[z]
 - Specified for voicing, pronounced as such
- Positions with neutralization: output [S]
 - Pronunciation determined by context

voiced / $_V$ and / $_C[+voi]$,

voiceless *elsewhere* (/ $_C[-voi]$,

/ $_$ _{Utterance} ...)

Cyclic approach: the main idea

- Derive alternations by changing /s/, /z/ $\rightarrow$ S at intermediate stage of derivation
 - E.g., /kaz/ $\rightarrow$ kaS, pronounced [kas], [kaz] according to context
- Other predictable contexts: also S
 - /VsIV/, /VzIV/ $\rightarrow$ [VSIV]



Cyclic approach

Word level delaryngealization

/káz/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. kás		*!	*	
b. káz		*!		
c. ➡ kás				*



Word level delaryngealization in cases without alternation

/oslu/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. ós.lu		*!		
b. óz.lu		*!	*	
c.  óS.lu				*

/ozlu/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. ós.lu		*!	*	
b. óz.lu		*!		
c.  óS.lu				*



Phrase level

Input to phrase level: /káS u.βért/

The phonetic realization will be derived
by markedness constraints, or phonetics:
[ká.zu.βért]



Incorporating prefixes

- Reminder: prefix-final stridents have predictable voicing, like word-final s,z
- Three potential URs for the prefix
 - /dəs/, /dəz/, /dəS/

The dangerous one

/dəs-ánim/	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. 🖐️ də.sá.nim				
b. də.zá.nim			*!	
c. 🐱 də.Sá.nim				*
d. 🐱 dəS.á.nim	*!			*

Adding Align-L

Word level

/dəs [ω .á.nim/	Align-L	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a.  dəS[ω .á.nim		*			*
b. də.[ω sá.nim	*!				

At the phrase level the constraint Align-L becomes irrelevant and resyllabification takes place.

Markedness constraints or phonetics take care of the final realization

 [də.zá.nim]

Catalan: a purely prosodic approach?

- The challenge: use prosodic word boundaries *alone* to replicate the effect of putting /s/ in a coda at a certain stage of derivation
- Pieces of an approach
 - Align-L
 - *Lar/_¬V **within the word**: no laryngeal features unless licensed by a following vowel within the word
 - Underspecified S acquires voicing values from context *phonetically* (Steriade 2001)



Prosodic words alone is not enough

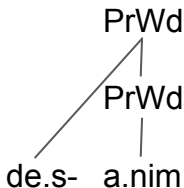
- Parallel approach, only one level of evaluation
- Constraints above won't select correct output

/dəs [ω .á.nim/	Align-L	Onset	*Lar] _σ	Ident (±voi)	Max(lar)
a. 🐱 də.S[ω á.nim					*!
b. 🖐 də.s[ω á.nim					
c. dəS[ω .á.nim		*!	*		*
d də.[ω sá.nim	*!				

- The problem: strident in question is not in a coda, so shouldn't lose voicing specification

Fixing this?

- Insight: although the strident in question is an onset, it is not within the same minimal PrWd as the following vowel



- Change *Lar to care about the PrWd domain?



Something that doesn't work

- *Lar/_¬V **within the PrWd**
- Although [ánim]_ω is a PrWd, *des-* is not a separate PrWd

/dəs [ω .á.nim/	Align-L	Onset	*Lar/_¬V	Ident (±voi)	Max(lar)
a. 🐱 də.S[ω á.nim					*
b. 📖 də.s[ω á.nim			we want a violation here, but we can't have it		
c. dəS[ω .á.nim		*!			*
d. də.[ω sá.nim	*!				



This doesn't help...

- *Lar/_¬V within the minimal PrWd

/dəs [ω .á.nim/	Align-L	Onset	*Lar/_¬V	Ident (±voi)	Max(lar)
a. 🐱 də.S[ω .á.nim					*
b. 🖐 də.s[ω .á.nim			we want a violation here, but we can't have it		
c. dəS[ω .á.nim		*!			*
d. də.[ω sá.nim	*!				



Is this legal?

- *Lar unless/_V **not separated by a PrWd boundary**
 - də.s[_ω á.nim
 - Violation: s[-voi] [_ω V
- This formulation seems to do the trick, but it is powerful and awkward

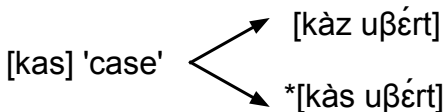
Catalan: a transderivational approach

Two challenges

- Word-final stridents devoice phrase-finally, but voice in phrases
 - Alternation, not BD identity?
- Prefix-final segments also voice /₋[+voi]

Voicing alternations as an identity effect

- The challenge



- Strategy: examine phonetic details beyond just voicing
 - Duration of fricative, surrounding segments
 - Perhaps other properties?

Voicing contrast in fricatives

/kasa/	IO-Ident ([±voi])/_V	*VsV	*D	IO-Ident ([±voi])
a.  kasə		*		
b. kazə	*!		*	*

/kaza/	IO-Ident ([±voi])/_V	*VsV	*D	IO-Ident ([±voi])
a. kasə	*!	*		*
b.  kazə			*	



Voicing contrasts in fricatives

- Voicing during frication
- Duration of frication (Hualde and Prieto 2014)
- VC ratio, etc? (Port and Dalby 1982)

[kasə]

- no voicing
- relatively longer
- shorter VC ratio
- ...

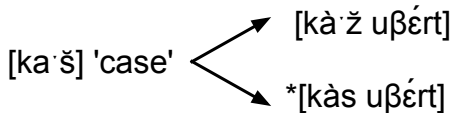
[kaːʒə]

- voicing
- relatively shorter
- longer VC ratio
- ...



Final consonants

- Assumption: gestural reduction in final position
- Prediction (yet to be verified): longer vowel, shorter frication: [ka·š]
- Idea: short final [š] differs from intervocalic [ž] in voicing, but a better durational match
 - Perhaps also longer preceding vowel



Isolation forms: final neutralization

/kas/	OO-Ident (C dur)	*VŠV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a.  kas ( [ka·š])					
b. kaz				*!	*

/kaz/	OO-Ident (C dur)	*VŠV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a.  kas ( [ka·š])					*
b. kaz				*!	



In phrases: voicing

- Assumption: high-ranked markedness against short or gesturally reduced š /V_V (*VšV)

/kaz ɔβɛrt/ (Base: [ka·š])	OO-Ident (C dur)	*VšV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a. ➡ ka·ž uβɛrt				*	
b. ka·š uβɛrt		*!	*		*
c. kas uβɛrt	*!		*		*

- Word-final voicing contrast still neutralized, but voiced output is favored /V_V because a better durational match to final [V·š]

In phrases: voicing

- Assumption: high-ranked markedness against short or gesturally reduced š /V_V (*VšV)

/kas ɔβɛrt/ (Base: [ka·š])	OO-Ident (C dur)	*VšV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a.  ka·ž uβɛrt			*	*	*
b. ka·š uβɛrt		*!			
c. kas uβɛrt	*!				

- Word-final voicing contrast still neutralized, but voiced output is favored /V_V because a better durational match to final [V·š]



Stridents in prefixes

- Reminder: /s/ at the end of prefixes undergoes contextual voicing
 - desinflar* [dəzimflá] ‘deflate’
 - desànim* [dəzánim] ‘discouragement’
- This is true of all s-final prefixes, so we can't rely on just saying that *des-* is underlyingly /dez-/
 - *des-*, *trans-*, *sus-*



Stridents in prefixes

- Elsewhere in the prefix: [s] initially, [z] intervocalically (really: between sonorants)
 - Initial: *semi-*, *sin-* *sots-*, *sub-*, *sus-*, *super-*, *supra-*
 - Intervocalic: *iso-*, *eso-*, *quasi-*
- One exception: *vice-* [bisə]



Voicing in roots vs. prefixes

- *cas obert* [kà.z uβért] ‘open case’ and *desinflar* [dəzimflá] ‘deflate’ both require voicing, but for different reasons
- Word-finally: no contrast in word-final position, and [VzV] is the more similar correspondent to [Vs#]
- In prefixes: no strident voicing contrast distribution is allophonic
 - Voiceless initially
 - Voiced intervocalically
 - Regressive voicing agreement /_C



Root vs. affix faithfulness

- Ident([±voi])/Rt >> *VsV >> Ident([±voi])

/kasa/	OO-Ident (C dur)	*VšV	IO-Ident([±voi]) /_V Root	*VsV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a. ➡ kasə				*			
b. kazə			*!		*	*	*

/des-infla/	OO-Ident (C dur)	*VšV	IO-Ident([±voi]) /_V Root	*VsV	IO-Ident ([±voi])/_V	*D	IO-Ident ([±voi])
a. ➡ dəzimfla					*	*	*
b. dəsimfla				*!			



Some remaining issues

- One exceptional prefix: *vice-* [bisə]
- Voicing contrast also in suffixes
 - -asə, -isə vs. -ozə

(Goal here is to illustrate the approach, so won't try to solve these details here)



Other "word-final" behavior of prefixes

- Prefix-final devoicing
 - /su**b**-adresa/ → [su**p**.əðresə]
 - BD-Ident approach for strident voicing won't cover this
- Cyclicity alone can't do it, either
- Prosody alone can't do it, either
- Combination of prosody and cyclicity?



s-aspiration in Buenos Aires vs. Granada



s-aspiration in Buenos Aires vs. Granada

Buenos Aires: [desetʃo] *deshecho* 'undone'

/des-etʃo/	*s]	Onset	Align-L
a.  de.setʃo			*
b. des.etʃo	*!	*	
c. deh.etʃo		*!	

Granada: [dehetʃo] *deshecho* 'undone'

/des-etʃo/	*s]	Align-L	Onset
a. de.setʃo		*!	
b. des.etʃo	*!		*
c.  deh.etʃo			*

Northern Italian s-voicing



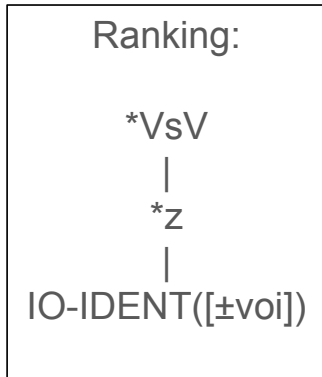
s~z allophony in Northern Italian

- [z] intervocalically

'kaza	house'
'vizo	face'
ri'zata	laughter'

- [s] initially (=elsewhere?)

'santo	saint
'sot:o	under
so'lare	solar



Voicing is blocked stem-initially

Examples from Nespor (1993)

a.	/isola/	izola	'island'
	/kaʊs-a-v-a/	kaʊzava	'he/she was causing'
	/famos-is:im-o/	famozis:imo	'very famous'
b.	/ri-sal-a-re/	risalare	're-salt'
	/porta-sigaret:a/	portasigaret:a	'cigarette case'
	/lavando-si/	lavandosi	'washing oneself'
	/la sera/	lasera	'the evening'
	/bɛl:a sera/	bɛl:asera	'beautiful evening'



Prosodic approach

ri-[s]*alare* 're-salt', *la* [s]*era* 'the evening', etc.

- Nespor and Vogel (1986)
 - Intervocalic voicing is restricted to PrWd
 - $*[...VsV...]_{PrWd}$
- Prosodic word = stem+suffixes
 - Prefixes, compounds, clitics, etc. in separate prosodic words
 - With complications...



A cloud on the horizon

- Preceding statement works well for compounds, clitics, other word boundaries
- Prefixes
 - No voicing after V-final prefixes

ri-[s]alare 're-salt', *pre-[s]elezione*
'preselection'

- But there is voicing in s-final prefixes

di[z]-uguale 'unequal'



Nespor and Vogel's complication

- Stipulate a prosodic difference
 - V-final prefixes form a separate PrWd
 - C-final prefixes are prosodified in the same word as the following stem
- Unappealing unless we can make this difference fall out from some independent factor
- Could this be due to Italian phonotactics?
 - No word-final consonants (*C#)



Phonotactics influences prosodification

- V-final prefixes can satisfy ALIGN-L without violating phonotactics

/CV-stem/	*C#	IO-DEP(V)	Align-L	*PrWd
a.  [CV][stem]				**
b. [CV.stem]			**!	*

- C-final prefixes prosodified with stem to avoid C-final PrWd

/CVC-stem/	*C#	IO-DEP(V)	Align-L	*PrWd
a. [CVC][stem]	*!			**
b.  [CVC.stem]			***	*
c. [CVCi][stem]		*!	*	*



But a problem: C misalignment

/CVC-stem/	*C#	IO-DEP(V)	Align-L	*PrWd
a. [CVC][stem]	*!			**
b. ☞ [CVC.stem]			***	*
c. [CVCi][stem]		*!	*	*
d. ☞ [CV][C-stem]			*	**

- Under the Alignment approach, better to have a slightly misaligned PrWd boundary than no PrWd boundary (so nearest one is far away)
- Categorical Align would solve this (but fails for German)



Prosodic approach: further predictions

- Prosodic approach to Northern Italian s-voicing ties (lack of) voicing to a PrWd boundary
- Prefixes that block voicing should act like separate PrWd's in all respects
- Phonotactic distributions, stress, etc.



A cyclic approach

- Cycle of evaluation prior to adding prefix
 - /sal-a-re/ → [salare]
 - /isol-a-re/ → [izolare]
- Subsequent cycle of evaluation maintains [s]
 - /ri-salare/ → [risalare] (s-voicing "turned off")



First cycle: s-voicing

- s~z allophony (Markedness >> Faithfulness)

/salare/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a.  salare			
b. zalare		*!	*

/isolare/	*V[+cont,-voi]V	*[+cont,-voi]	IO-IDENT([±voi])
a. isolare	*!		
b.  izolare		*	*



Second cycle: prefixation

- Prefix /ri-/ + output of first cycle [salare]
- s~z allophony must be turned off to preserve output of first cycle (Faithfulness >> Markedness)

/ri-salare/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,-voi]
a.  risalare		*	
b. rizalare	*!		*



Cyclic approach: further predictions

- Since faithfulness is ranked high when the prefix is added, /s/ vs. /z/ contrasts in prefixes should be preserved
- This is wrong in two ways
 - Prefix-final /s/ voices: *di[z]-uguale* 'unequal'
 - Prefix-medial stridents are always voiced



Prefix-final /s/

- Prefixes that end in /s/ voice before a vowel
 - *di*[z]-*uguale* 'unequal'
- Consistent across /s/-final prefixes
 - *dis-*, *es-*, *tras-*, *cis-*, *bis-*, *mis-*
- These s~z alternations require *VsV >>
IO-Ident([±voi])



Prefix-medial s

- Medial VsV in prefixes always show voicing, as well
 - izo- 'iso-', ezo- 'exo-', eza- 'hexa-', kwazi- 'quasi-'
- Also scientific formatives
 - krizo- 'chryso-' (gold), nazo- 'naso-', nozo- 'noso-' (disease)
- ROTB: this predictability requires *VsV >> IO-Ident([±voi])



Consequences for stratal re-ranking

- The fact that /s/ in prefixes always respects intervocalic voicing is a problem for an approach that relies on reranking $\text{Ident}([\pm\text{voi}])$ above $*\text{VsV}$
- Still seems necessary to assume that initial [s] from first cycle has some special property that protects it
 - E.g., at a prosodic boundary



A transderivational approach

- Kenstowicz (1996): block voicing in *ri-[s]alare* 're-salt' by faithfulness to *[s]alare* 'salt'

/salare/	BD-Ident ([±voi])	*VsV	*z	IO-IDENT ([±voi])
a.  salare				
b. zalare			*!	*

/ri-salare/	BD-Ident ([±voi])	*VsV	*z	IO-IDENT ([±voi])
a. risalare				
b.  rizalare	*!		*!	*

Transderivational approach: predictions

- BD-Ident([±voi]) can only be violated if there is a free-standing (output) base form
 - *ri-salare* ← *salare*
- Bound forms: no base, no blocking
 - *re-[z]istente* 'resistent' (no **sistente*)
- Baroni (2001) points out several bound stems that usually or always resist voicing



Bound stems that resist voicing

Baroni (2001)

ante-[s]ignano	'forerunner'	*signano
a-[s]intoto	'asymptote'	*sintoto
poli-[s]emo	'polysemous'	*semo
poli-[s]indeto	'polysyndeton'	*sindeto
pre-[s]idio	'garrison'	*sidio

also: *dino[s]auro*



Bound stems that resist voicing

- Transderivational approach has no way to block voicing in these forms
- As with cyclic approach, must be augmented somehow
 - Prosodic boundaries?
 - Learned exceptions?



References

- ALBER, BIRGIT. 2001. Regional Variation and Edges: Glottal Stop Epenthesis and Dissimilation in Standard and Southern Varieties of German. *Zeitschrift für Sprachwissenschaft* 20.
- HUALDE, JOSÉ IGNACIO.; and PILAR PRIETO. 2014. Lenition of Intervocalic Alveolar Fricatives in Catalan and Spanish. *Phonetica* 71.109–127.
- PORT, ROBERT F.; and JONATHAN DALBY. 1982. Consonant/vowel ratio as a cue for voicing in English. *Perception & Psychophysics* 32.141–152.